TECHNICAL OVERVIEW / 0.5.0 DEVELOPMENT SOURCE

Khmer Easy Tools

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Prepare text. Tag words. Search locally. Read documents.

81,005
dictionary entries

239 KiB
bundled research POS model

0
core runtime dependencies

01 Everyday tools

Text preparation

Audit + reversible cleanup
Digits + lossless source spans
Sentences + pattern extraction

Words and retrieval

Segmentation + Khmer POS
Local BM25 keyword search
Passages with source citations

Documents and setup

Image / PDF OCR
Lines, blocks + word boxes
One-call process + doctor

02 Small core, explicit optional backends

Text → **Prepare / segment / tag / retrieve**
Offline, standard-library inference
→ **Results**
with spans

Optional OCR: Pillow + PDFium + Tesseract with Khmer language data.

Optional translation / generated answers: explicitly configure a service or local model.

03 Evidence with visible limits

Word segmentation — **78.46% F1**
khPOS OPEN; 1,000 sentences / 10,778 tokens

Native POS, gold word boundaries — **95.94%**
Purged khPOS CLOSE; 1,000 sentences / 10,397 tokens

OCR character error rate — **11.43% CER**
40 KHOB images = 10 documents x 4 variants; lower is better

Windows / macOS / Linux: fresh native installations passed.

Each OS: 203 core tests + 217 extras tests passed; 19 / 5 skipped.
Actual OCR + native POS executed; 2,000 compact-POS sentences replayed per OS.

Native joint word/POS F1: 87.78%; mapped public POS: 94.23% at gold boundaries.

Role hints are heuristic. OCR is Tesseract-based; general sub-10% CER is not established.

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MIT software; CC BY-NC-SA 4.0 trained POS weights. Source: TAGGER, IMPROVEMENTS and VALIDATION guides.